Job Market & Unemployment Trends

Professional analytical report on 1,000 observations across 20 U.S. regions (2023–2025) — exploring how unemployment varies by geography, education, and demographics, and how these factors shape job offer volume and nature.

PYTHON → SQL → POWER BI

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Dataset & Pipeline Overview

Dataset

- 1,000 observations | 20 U.S. regions
- Period: 2023–2025
- 8 raw columns → 12 final columns
- Key variables: unemployment rate, job offers, market tension, region, age group, education level, skill family

Pipeline

01

Python — EDA & Cleaning

IQR outlier capping, datetime normalization, 8 new engineered features

02

SQL — MySQL Workbench

10 key analytical queries via SQLAlchemy + pymysql

03

Power BI — Dashboard

4-page interactive dashboard with maps, charts, and matrices

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Executive Snapshot

8.63%

Avg. Unemployment

Global average across all 20 regions

2M

Total Job Offers

Across the full 2023–2025 period

313

Market Tension

Average market tension index

-0.01%

Annual Variation

Positive trend: improvement underway in 2025

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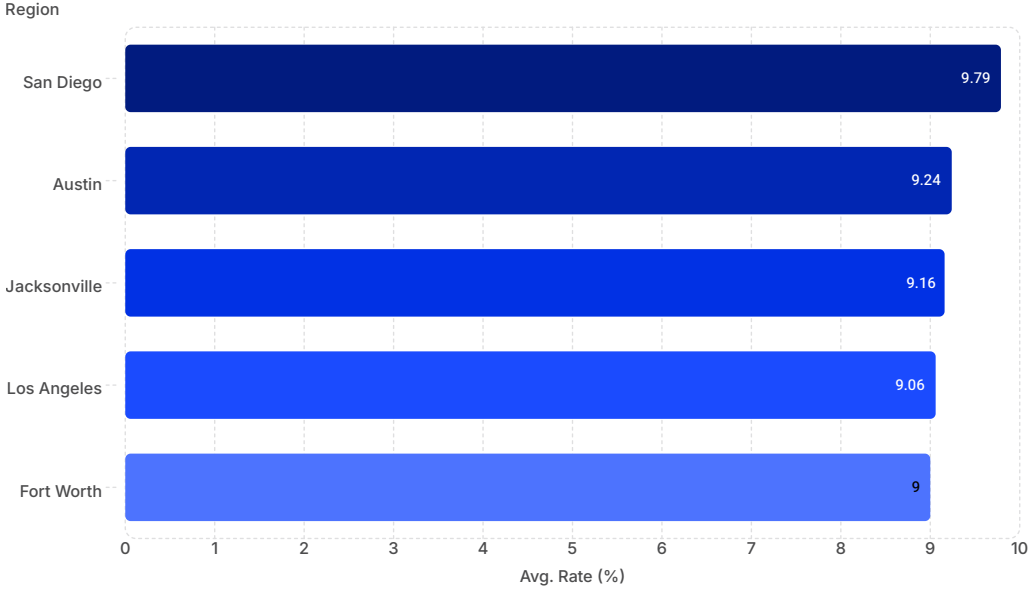
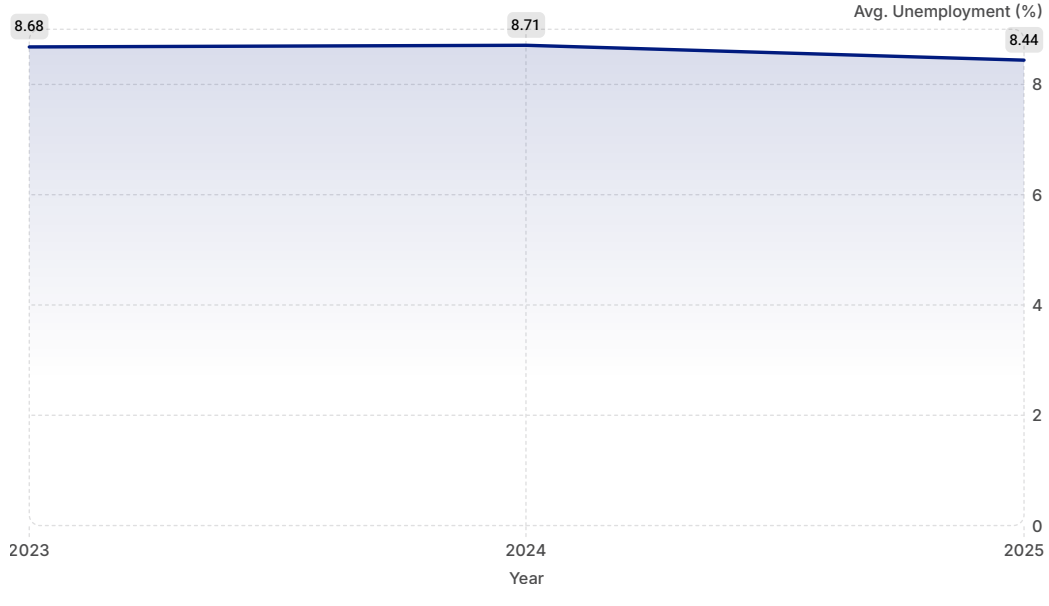
Regional Unemployment — Top 5 & Trends

Highest Unemployment Rates



Annual Evolution



2024 was the peak year; 2025 marks a positive turning point.

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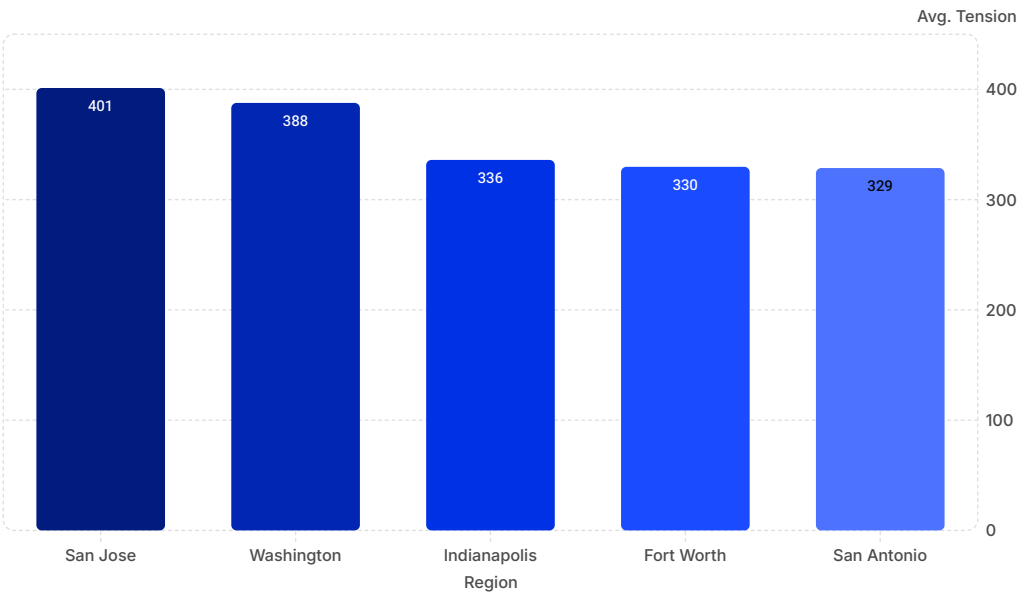
Market Tension & Geographic Vulnerability

Top 5 Market Tension Index



Double Vulnerability Check (Q3)

No region simultaneously combines **high unemployment** and **low job offers**.

✅ Positive signal: no zone of absolute double vulnerability detected across all 20 regions.

High market tension in San Jose and Washington reflects strong demand relative to supply — a competitive, not distressed, market.

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Demographics & Education Impact

Unemployment by Age Group

Active (30–45)

8.73% unemployment | 2,556 avg. offers Highest rate despite most offers — mismatch problem

Young (<30)

8.54% unemployment | 2,375 avg. offers

Senior (>45)

8.45% unemployment | 2,455 avg. offers

Unemployment by Education Level

Low Education

8.45% unemployment | 2,535 avg. offers

High Education

8.58% unemployment | 2,570 avg. offers Counter-intuitive: more offers, not lowest unemployment

Medium Education

8.69% unemployment | 2,419 avg. offers

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Skills & Sectors — Tech & Data Dominates

2.2M

Tech & Data Offers

15× the volume of the second-largest sector

136K

Others

Second largest skill family by total offers

112K

Marketing & Com

Lowest unemployment by sector: **8.0%**

73K

Trade & Sales

Highest unemployment by sector: **9.2%**

Top region × skill couples: **Washington + Tech & Data** (143,120 offers), **Austin** (133,236), **Chicago** (131,983). All top 10 couples dominated by Tech & Data.

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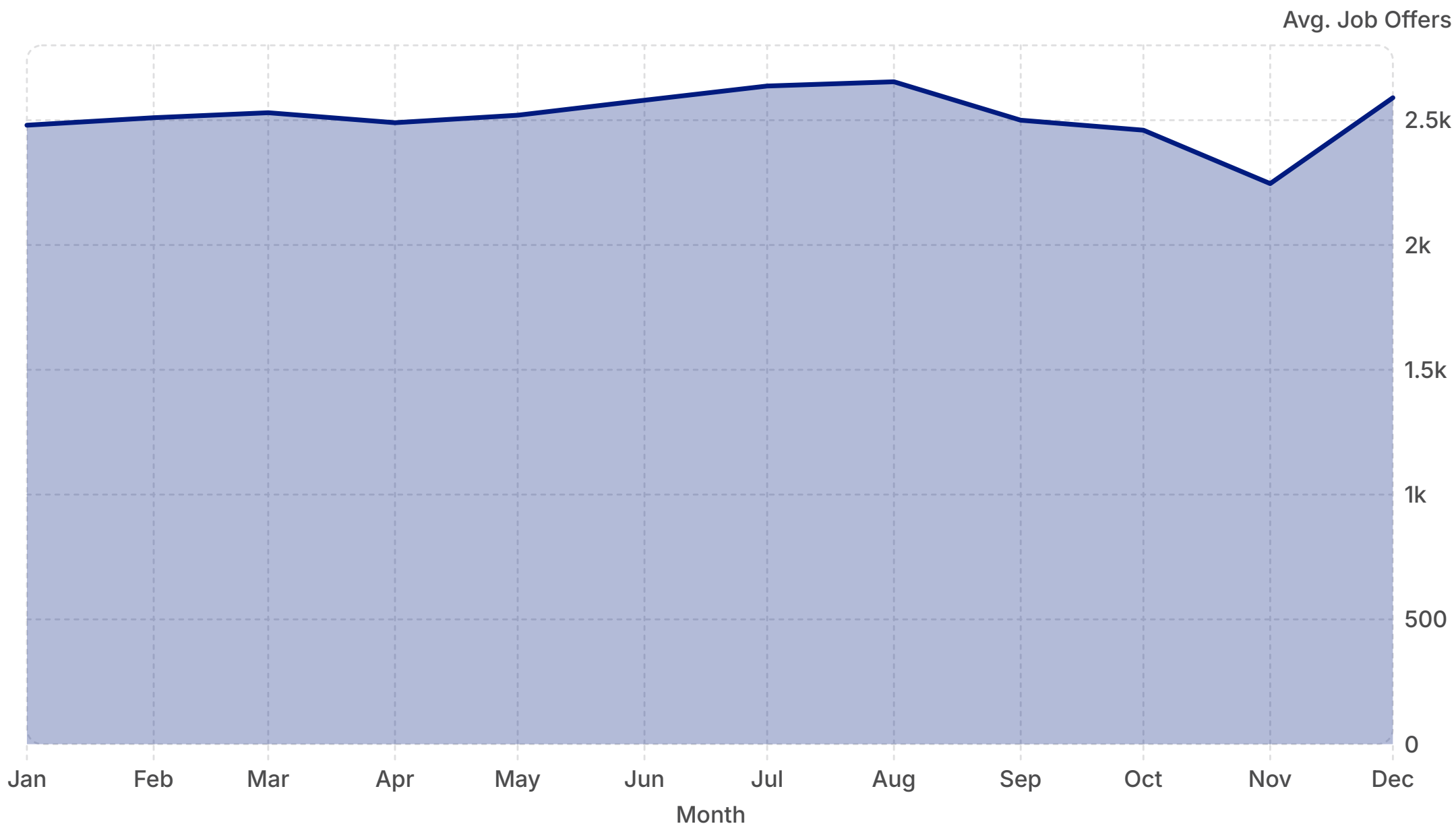
Recruitment Seasonality



📈 Peak 1: August — 2,654 avg. offers | Peak 2: July — 2,637 avg. offers

📉 Trough: November — 2,246 avg. offers. Launch placement campaigns in June–July and October.

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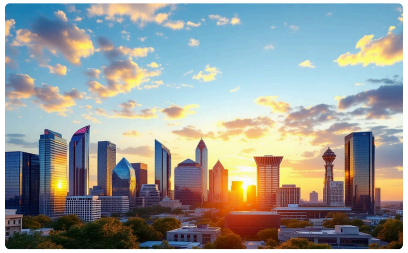
Low-Unemployment Region Profile



Austin

Unemployment: **~2.0%** Profile: Tech & Data + Medium/High Education + Active/Young workforce

San Antonio

Unemployment: **~2.1%** Profile: Tech & Data + Education + Active workforce accompaniment

San Diego

Unemployment: **~2.35%** Profile: Tech & Data + High Education + Young/Active demographic

Common model: **Tech & Data ecosystem + Medium/High education + Active/Young demographic** — reproducible for vulnerable zones.

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Business Recommendations

Territorial Intervention

Deploy active placement units in San Diego (9.79%), Austin (9.24%), Jacksonville (9.16%) with local Tech & Data employers

Digital Training

Fund Data/Cloud/Cybersecurity bootcamps in high-unemployment, high-tension regions

Reskill Ages 30–45

Short requalification tracks (3–6 months) targeting Tech & Data for the most affected age group

Anticipate Peaks

Launch employer/candidate matching in June and October to capture August and December hiring surges

Replicate Top Models

Transpose Austin & San Antonio's formula (Tech + education + active support) to vulnerable zones

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